



RESTORATIONESSENTIALS

Factory-Correct Restoration Reference

Coyote Conversion Guide — Mustang 1965–1970

Model Year 2026

Recipient cars	Mustang coupe/fastback/convertible (1965–70)
Donor engines	Coyote Gen 1–3 donor-first · crate + control pack · Gen 4 verify-first
Highlights	Tower decision graded permanent; PATS/control-pack chapter; worksheets
Price	\$29.95 — PDF download at checkout

Every specification in this guide was checked against factory literature, marque references, period road tests, and documented survivor cars. Where sources disagree or the record is silent, this guide says so rather than guessing.

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Chapter 1 — The Decision: Donor, Crate, or Control Pack (Read This First)

Every Coyote swap into a 1965–70 Mustang is decided by three purchases made in the wrong order by most first-timers: the engine, the front suspension, and the electronics. Buy the engine first without the other two and you own a wide dual-overhead-cam V8 that fits a narrow shock-tower bay only after structural decisions you have not priced yet, and a computer that — from a donor car — will not run outside its original car without specialist help. This chapter puts the whole bill on the table first.

Which Coyote. Ford published the Mustang GT generations as: Gen 1 (2011–14) at 412 hp / 390 lb-ft, rising to 420 hp for 2013–14; Gen 2 (2015–17) at 435 hp / 400 lb-ft; Gen 3 (2018 and later) at 460 hp / 420 lb-ft, adding combined direct and port injection. Newer Gen 4 engines raise the numbers again and narrow the swap-parts record this guide stands on, so the build chapters pin Gen 1 through Gen 3 and treat Gen 4 as verify-before-you-buy. Those are Ford's published ratings, not this guide's promises — a salvage engine makes what its condition makes. One warning printed early because it costs people real money: an F-150 Coyote is not a Mustang Coyote deal in disguise. The truck engines are rated and tuned lower, their intake and accessory packaging differ, and Ford Performance control packs explicitly exclude them.

- **Donor-first (this guide's default):** a Mustang GT salvage engine of a known generation, audited in Chapter 2 (compression, accessory completeness, harness and PCM status). Cheapest entry; the electronics decision below is where the donor path spends its savings back if you let it.
- **Crate:** a new Ford Performance Coyote with a matched control pack. Known quantity, real money, warranty on the engine. The tower work, suspension, fuel, cooling, and transmission bills below are identical either way — the crate buys certainty about the engine, not an easier car.
- **The electronics are a decision, not an accessory:** Ford Performance control pack (generation- and transmission-specific), specialist-reflashed donor PCM with the anti-theft (PATS) system defeated, or aftermarket standalone. Chapter 5 prices and grades all three. Skipping this decision until the engine is in the car is how Coyote swaps become garage art.



The line items first-timers miss (planning ranges, not promises — prices move): the shock-tower decision and the front-suspension system it implies (the single biggest line after the engine); mounts and the swap oil pan matched to that suspension; accessory-drive relocation for the alternator and A/C compressor; headers chosen for your suspension, not your engine; the control pack or its alternatives; a return-style EFI fuel system end to end (a second, higher-pressure story on Gen 3); electric fans and the radiator package; transmission and its tunnel answer (Chapter 3); driveshaft measurement; gauge solutions. Restoring the car first? The RestoreEssentials restoration guides for 1964½/1965 through 1970 Mustang are the companion volumes — restore it with those, convert it with this one. And one honesty note this guide will not soften: if your car is a rare or numbers-matching example, Chapter 3's permanent modifications are a reason to convert a different car. The guide says so in the chapter where it matters.

Note. If you only have \$X: spend it in this order — (1) the front-suspension/tower decision made once, correctly, (2) the control pack or its proven alternative, (3) the fuel system done once at specification, (4) cooling, (5) cosmetics. A Coyote that starts, cools, and steers beats a polished engine bay that does none of the three.

Chapter 2 — What You Start With: The 1965–70 Mustang Baseline

Inspect the recipient car before the donor engine gets purchased. Pass/fail criteria, not vibes:

- **Towers and rails:** the front shock towers and frame rails are structure on a unibody, and they are also the rust-prone metal of this era. Tower rust bubbles, rail section loss, or prior collision repair in the front clip = structural repair first (the restoration guides and body shops own that work), because Chapter 3 will ask this metal to change jobs. A car whose towers are rotten is a candidate for the full front-clip suspension path; a car whose towers are perfect is a candidate for thinking twice (next bullet).
- **The car's identity:** a common coupe is the honest recipient. A documented K-code, a Shelby, a numbers-matching big-block car — tower deletion erases value that a restoration preserves. This guide names that plainly and points at the restoration guide. Nobody gets to be surprised by it later.
- **Steering and brakes as found:** note the existing steering (manual or power box) and brake system honestly. The front-suspension decision replaces most of the steering story; the brakes must be planned to the new engine's weight and speed, and Chapter 13 enforces a stopping standard before road speed.
- **Tunnel:** measure what you have (Worksheet B). The tunnel was drawn around Toploader and C4-era cases; the six-speed chapter (Chapter 3) is decided by your measurement, not by enthusiasm.
- **Ride height:** record it as the car will be driven. Driveshaft length and pinion angle are measured at final height (Worksheet A).
- **The donor audit** (family discipline): generation confirmed in writing (Chapter 1's table against the donor's year and source car), compression or a running demonstration, and a signed accessory inventory — alternator, compressor, starter, throttle body and pedal, intake hardware, PCM status. Mustang GT vs F-150 provenance is verified here, not discovered at the parts counter.

Chapter 3 — Fitment: The Wide Engine in the Narrow Bay

The one-sentence version: a dual-overhead-cam 5.0 does not fit between 1965–70 shock towers as they stand, and every honest path through that fact is a permanent-modification decision made on paper first. Two schools, both documented in the swap record:

- **School 1 — towers out.** An aftermarket front suspension and crossmember (Mustang II-style or modern strut/coilover clips, bolt-in or weld-in) removes the towers entirely, bringing rack-and-pinion steering, modern



geometry, and the engine room the Coyote needs. The most complete answer and the most welding: chassis prep and alignment are part of the price, and it is irreversible. PERMANENT — graded that way here and in Chapter 9's gotchas.

- **School 2 — towers trimmed.** Strut-based systems that retain the tower tops and trim the rest back for clearance. Less invasive on paper; still cutting and welding structure, still permanent, and the clearance result is measured on your car — tower spacing varies across 1965–70 (early cars are the tightest), so a trim that clears on a '69 is not promised on a '65.
- **What there is not:** an honest bolt-in path that leaves the towers untouched. Any seller promising one is selling the surprise separately.

Oil pan follows the suspension choice. Swap pans for Coyote muscle-car chassis exist in front-sump form (the Canton 15-738 with its matching 15-739 pickup is one documented example, sized for stock muscle-car-era Ford chassis once the engine itself clears), and fabricated road-race pans come from the usual makers. The pan is selected after the crossmember is chosen, and its clearance to that crossmember is MEASURED ON YOUR CAR at mock-up (Worksheet B) — a pan table cannot know your crossmember.

Accessory drive is the second tight fit. The Coyote's front drive puts the alternator low, and modern donors used electric steering assistance — so a hydraulic power-steering pump for your classic steering (or the rack in School 1) is an added bracket problem, and A/C compressor placement competes with towers and rails that no longer exist in factory positions. Relocation and low-mount bracket kits exist precisely because of this; the fit is verified complete — every pulley and bracket — at mock-up, against your towers or your clip. Buying brackets before the suspension decision is buying twice.

Transmission and tunnel. The MT82 six-speed manual and the 6R80/10R80 automatics are larger cases than anything Ford planned for this tunnel, and tunnel enlargement is a commonly documented step for the six-speed swaps (crossmembers exist for the 6R80/MT82 pairing; the floor still gets measured, Worksheet B). The smaller five-speed (TKO/TKX-class) manual conversions are the less-invasive path and the reason many classic builds choose them. Whichever you choose: transmission choice, crossmember, and tunnel are one decision, made before the engine's final position is fixed — and tunnel work on a unibody is permanent, so it is chosen last, knowingly.

Chapter 4 — Know Your Generation: Gen 1, Gen 2, Gen 3

The Coyote badge covers three engines a swapper must tell apart, because they differ in induction, fuel hardware, and control-pack pairing. Ford's published Mustang GT ratings: Gen 1 (2011–14) at 412 hp and 390 lb-ft, rising to 420 hp for 2013–14, port injection only, 11.0:1 compression; Gen 2 (2015–17) at 435 hp and 400 lb-ft, still port-injected with revised intake, cams, and heads; Gen 3 (2018 and later) at 460 hp and 420 lb-ft with combined direct and port injection and 12.0:1 compression. Gen 4 (2024 and later) raises output again; this guide treats it as verify-before-you-buy because the classic-swap record and control-pack coverage for it were still maturing when this guide was written.

- **Identification is a purchase step, not trivia:** donor year and source vehicle, intake and throttle-body configuration, and the direct-injection hardware on Gen 3 (engine-mounted high-pressure pump and its lines) distinguish the generations in the salvage yard. F-150 Coyotes of every generation are rated and tuned lower and are excluded from Ford Performance control packs — Chapter 2's provenance audit is where this gets signed, before money moves.
- **What the generation changes downstream:** Gen 3's dual injection makes Chapter 6's fuel supply a two-pressure problem; control packs are generation- and transmission-specific and do not cross over (Chapter 5); and the tuning brief in Chapter 9 starts from a different factory calibration in each generation. Every later chapter assumes the generation written in your Worksheet C. If you cannot name yours, stop and name it.

Chapter 5 — Wiring and Controls: The Control-Pack Decision

This is the chapter the Coyote is famous for, and it is a purchasing decision, not a wiring weekend. Three roads:

- **Ford Performance control pack (the documented default):** generation- and transmission-specific kits (the M-6017 series: Gen 1 M-6017-A504VA class; Gen 2 manual M-6017-504V with the 6R80-automatic sibling M-6017-M50A; Gen 3 manual M-6017-M50B; Gen 4X packs for the newest crates). Published contents: a PCM with Ford Performance calibration; the electronic throttle (drive-by-wire) pedal; a unique control-pack harness that replaces the stock body harness for the engine; an OBD-II diagnostic port; a power distribution module; air box and inlet tube; MAF sensor; upper and lower radiator hoses; and HEGO sensors. Two published conditions travel with it: the calibration requires a return-style fuel system (it will not run a returnless setup — Chapter 6 plans accordingly), and the pack is built for Mustang GT engines — it explicitly excludes F-150 Coyotes. A control-pack PCM installed in a production Mustang produces a no-start by design; it is a swap calibration, and that is a feature when it is in your 1968 coupe.
- **Donor PCM, specialist-reflashed:** the donor car's PCM with its Passive Anti-Theft System (PATS) defeated by a specialist service. Contemporary swap coverage is blunt that PATS resisted the simple handheld-tuner defeats that worked on earlier Fords, which is why this road is a send-it-out service with references, not a Saturday with a laptop. Budget the service, verify the tuner supports your exact generation, and keep the donor's pedal and throttle body matched to that PCM.
- **Aftermarket standalone:** Holley/AEM/FAST-class systems exist for the Coyote. Published swap guidance notes the Ford Performance pack was the option that retained full Ti-VCT variable cam timing, while alternatives historically locked the cam phasers (fixed timing) or layered over the factory controls. Standalone support moves; verify current Ti-VCT capability in writing before purchase rather than accepting either this guide's summary or a forum's. Choose standalone for features (forced induction management, data logging), not to dodge the PATS conversation.

Either way: the drive-by-wire pedal gets mocked before it is mounted (the PCM reads pedal angle, not effort; a bad mount is a forever drivability complaint), the OBD-II port goes where you can reach it with the door closed, and gauge strategy is chosen now — classic senders adapted where the Coyote provides ports, pulse scaling for the tachometer, and a speedometer solution (electronic conversion, cable-drive box, or GPS) against the modern transmission's speed output.

Note. Safety: battery disconnected for every harness step. ECU grounds to clean engine metal, never to paint. And an emissions note that belongs in this chapter because the control-pack listings print it: Ford Performance rates these packs for competition/off-highway use; what your state allows on the road is Chapter 9's and your state's conversation, stated — never guaranteed.

Chapter 6 — The Fuel System: Return-Style, By Generation

Begin with the control pack's published condition: a return-style fuel system is required; returnless need not apply. From there, generation decides the hardware:

- **Gen 1 and Gen 2 (port injection):** a conventional EFI supply story at modern pressure — an in-tank pump module or baffled EFI tank (the carbureted Mustang tank has no sump and starves an external pump on corners), EFI-rated lines, filtration per the pump and injector makers' published requirements, a regulator on the return, and a deliberately vented tank with a rollover path. The exact pressure target comes from your control pack or PCM documentation for the generation you bought; the guide prints the method, the documentation prints the number.

- **Gen 3 (direct + port):** everything above, plus respect for the engine-mounted high-pressure pump it feeds. The low-pressure side must hold the DI pump's inlet specification at full load — a marginal pump that 'runs fine' at cruise is a full-throttle misfire with a fuel receipt. DI lines and fittings are specialized hardware, not hose-clamp territory; where the documentation specifies a fitting or a torque, it is a stop-and-comply item.
- **Either generation:** no carb-era rubber anywhere in the pressure path, lines routed away from exhaust heat, and a first-prime inspection with the coils unplugged before any cranking (Chapter 13's discipline).

Note. Safety, plainly: pressurized gasoline finds loose fittings at first prime, and Gen 3 adds a high-pressure side that punishes improvisation. Prime, inspect with a light and a dry hand, extinguisher in reach, no road test with any weep — there is no small gasoline leak.

Chapter 7 — Cooling: Electric Fans Are Not Optional

The Coyote's water pump carries no provision for a mechanical fan — electric fans are mandatory, which makes cooling an electrical decision (Chapter 5's outputs, the alternator's capacity) as much as an airflow one. The factory installation discipline travels with the swap: the system is conventional-flow with an inline remote thermostat housing that contains the bypass, and the coolant fill point must be the highest point in the system — where the swap packaging makes that impossible, a remote surge or supply tank is mandatory, not a styling choice, because air that cannot be purged becomes a hot right cylinder head. The heater-hose detail belongs here too: where no heater is fitted, the supply and return are spliced with the documented restrictor provision so the right head still purges and flows. Radiator and shroud are sized for idle-in-traffic with A/C if the car will have it, and the shroud is measured, not assumed. Overheating diagnosis order: fill/air discipline first, fan triggers second, shroud sealing third, capacity last.

Chapter 8 — Exhaust: Headers Are a Suspension Decision

On this swap, the header question cannot be answered until Chapter 3 is answered: a header built for a tower-delete clip lives in space that a trimmed-tower car does not have, and the steering it must clear is different in the two schools. Swap headers exist for the aftermarket front clips and for the trimmed-tower cars from the usual makers; buy by chassis and measured clearance (Worksheet B) with the steering at full lock, both ways. The clearance-first fallback is manifolds — Ford Performance's cast street-rod manifolds (M-9430-SR50 class, related to the truck pieces) are the documented example — and this family's honesty rule applies word for word: manifolds lose some power and clear everything, and on a street-driven classic that trade is often right. Ground clearance on a lowered car is MEASURED ON YOUR CAR at final ride height; collectors meet speed bumps in person.

Chapter 9 — ECU and Tuning Decisions

On this engine the practical tree is short:

- **Control-pack calibration, stock engine:** the documented default. It runs the engine it was built for with its cam timing intact; modified engines and supercharged variants require custom calibration (the listings say so plainly).
- **Control-pack or donor PCM, custom tune:** required when the airflow story changes — cams, forced induction, intake changes beyond the pack's brief. A mail-order file cannot know your cam.
- **Standalone:** chosen for features (Chapter 4), with current Ti-VCT support verified in writing first.
- **The no-start checklist before it is an emergency:** which PCM, which anti-theft status, matched pedal and throttle body, return-style fuel confirmed. Every Coyote no-start story is one of those four wearing a disguise



(Chapter 12 assumes all four answered).

What a tune cannot fix: an unvented cooling system (Chapter 7), a low-pressure fuel supply (Chapter 6), or a driveshaft angle (Worksheet A). The tune is the last system finished, not the first excuse used.

Chapter 10 — The Gotchas

Each gotcha in the family's fixed format: *What happens / Why it happens / How to see it coming / The fix, in order of cost*. Every category the family grades is here; where a category is clear on this body, it says so in one line.

- **The towers.** *What:* the Coyote's cam covers meet the shock towers. *Why:* a wide DOHC engine in a narrow pushrod bay. *See it:* Chapter 3, on paper, before the engine purchase. *Fix, in cost order:* there is no cheap order — tower trim with a strut system, or tower deletion with a front clip. Both PERMANENT; both chosen knowingly. The fix for buying first and measuring later does not exist, which is why the chapter order in this guide is what it is.
- **The F-150 engine that isn't.** *What:* the bargain donor will not run on Mustang GT control packs, and its intake, cams, and rating are truck parts. *Why:* a Coyote badge is not a specification. *See it:* Chapter 2's provenance audit. *Fix:* verify before money; after money, the fix is a different electronics plan priced honestly.
- **PATS, the silent no-start.** *What:* a donor-PCM engine cranks healthily and never fires. *Why:* the passive anti-theft system does not recognize your 1968 coupe as its car. *See it:* Chapter 5's decision made at purchase time. *Fix, in cost order:* specialist PATS-defeat reflash with references; control pack; standalone. Discovering it at first start costs the same money plus a month.
- **Returnless fuel under a return-style calibration.** *What:* the control-pack PCM commands a system the car doesn't have. *Why:* the published requirement was read as a suggestion. *See it:* Chapter 6. *Fix:* build the return system the calibration requires; there is no settings menu for this.
- **Accessory drive vs what's left of the bay.** *What:* alternator or compressor meets tower, rail, or clip. *Why:* the drive was packaged for an S550, and your steering pump was not in its plans. *See it:* complete drive on the engine at mock-up (Chapter 3). *Fix:* relocation/low-mount bracket kits matched to your suspension choice — bought after Chapter 3, not before.
- **The six-speed vs the tunnel.** *What:* MT82/6R80/10R80 cases meet a Toploader-era tunnel. *Why:* forty years of case growth. *See it:* Worksheet B dimensions against the transmission's published dimensions. *Fix:* the smaller five-speed path; tunnel enlargement, permanent, chosen last and knowingly.
- **Air that will not leave the cooling system.** *What:* a hot right head behind a calm gauge. *Why:* the fill point is not the highest point, or the no-heater hose provision was skipped. *See it:* Chapter 7's discipline at fill time. *Fix:* surge tank and correct hose routing; both cheaper than a head.
- **Throttle-pedal geometry (drive-by-wire).** *What:* a pedal mounted at the wrong angle drives lazy or touchy forever. *Why:* the PCM reads angle, not effort. *See it:* mocked mount, full travel verified. *Fix:* reposition; no tune cures geometry.
- **Donor accessory completeness.** *What:* the 'complete pullout' lacks the pedal, the throttle body, the intake hardware, or the PCM. *Why:* 'complete' is a sales word. *See it:* Chapter 2's signed checklist. *Fix:* inside the negotiation, never after it.
- **Driveshaft angle after ride-height change.** *What:* vibration that survives balancing. *Why:* the stance changed the working angles. *See it:* Worksheet A at final height. *Fix:* angle correction per the drivetrain guides; a shaft built to your measurement, never a forum's number.
- **Checked — clear on this body:** oil pan vs steering linkage in the F-body sense (your steering is in front of the crossmember or is a rack by now — the pan check that applies is pan-to-crossmember, Worksheet B); steam ports (the Coyote's air-purge discipline is Chapter 7 and graded there).



Chapter 11 — Tips and Tricks

- **Make the tower decision with the car, not the catalog:** towers on the hoist, engine on the stand, tape measure between them. The decision this swap turns on is a measurement.
- **Plan on three engine installations:** mock-up, clearance-check, final. With towers, accessories, and headers all voting, the builders who do it once planned three and got lucky twice.
- **Buy the electronics with the engine, as one negotiation.** Pedal, throttle body, PCM status, and harness completeness are worth more together than scattered across six later orders.
- **Bag and tag the donor teardown.** The connector you cut 'because it looked unused' is the one the PCM wants.
- **Photograph every routing decision.** Heat and chafe are patient enemies, and the photo is the map at the first repair.
- **Chase threads before torque values.** Sixty-year-old unibody threads lie to torque wrenches — and several of them are about to hold a front clip.

Chapter 12 — Special Tools

Must-have: an engine hoist with a leveler (a wide DOHC engine goes in at an angle between towers or clip rails; the leveler is the tool), rated jack stands, welding capability or a welder you trust for the Chapter 3 structural work — named here as a skill-and-equipment gate, not a shopping item — a fuel-pressure gauge matched to your generation's documentation (Chapter 6), a multimeter, and thread chasers. **Rent or borrow:** the transmission jack, the harmonic-balancer puller/installer, an engine stand rated for a dressed Coyote, the flare tool if hard lines are being bent. **Nice:** a lift — it converts Chapter 13's under-car checks from an afternoon to an hour. No tool is listed here without the chapter that uses it.

Chapter 13 — First Start and Shakedown

- **Before cranking:** fuel system primed and leak-checked with coils unplugged (Chapter 6); cooling filled with the Chapter 7 air discipline verified; oil system per the engine's procedure; the four no-start answers confirmed (PCM, anti-theft status, matched pedal/throttle, return-style fuel); extinguisher in reach.
- **First start:** one person at the key, one at the bay with a light and the OBD-II reader. Short cranks; oil pressure and fuel pressure are the gauges that matter today.
- **First heat cycle:** fan triggers verified against a real temperature; the right head's purge re-checked; fasteners torqued in structural work re-verified per their documentation.
- **50 miles:** leaks, belts, harness chafe, driveline report (Worksheet A if it vibrates), tower/clip weld inspection after heat cycles. **500 miles:** re-torque list, fluid levels, alignment after springs settle — and Chapter 9's honest conversation about what the tune still owes you.

Note. Safety, plainly: do not road-test until brakes are verified at walking pace, then neighborhood pace — a swapped car has different weight, different brakes (planned in Chapter 2), and a different pedal story. Steering and suspension fasteners from the Chapter 3 work are single-use-critical: they are torqued per their manufacturers' documentation on inspected metal, never 'snug by feel' on the parts that steer the car.



Appendix: Worksheets (Fill These In On YOUR Car)

Worksheet A — Driveline. Ride height (four corners, as the car will be driven): _____. Transmission output angle: _____. Pinion angle: _____. Working angles computed: _____. Shaft length, center-to-center at ride height: _____. Rear axle fitted (8-inch / 9-inch / other): _____. (The drivetrain family's rear-end and shaft guides own the strength and angle targets; these blanks own your car.)

Worksheet B — The permanent-modification record. Tower path chosen (trim / delete / clip, with system name): _____. Tower-to-cam-cover clearance at mock-up, L/R: ____ / _____. Pan-to-crossmember clearance: _____. Alternator and compressor clearance to towers/rails: _____. Header-to-steering clearance at full lock L/R: ____ / _____. Transmission chosen: _____. Tunnel clearance at closest point; floor modified (Y/N, permanent): _____. Write the permanent items down — the next owner deserves your honesty as much as you do.

Worksheet C — Parts-total ledger. Engine/donor: _____. Front suspension/tower work: _____. Mounts and oil pan/pickup: _____. Accessory drive and brackets: _____. Control pack / PCM service / standalone: _____. Fuel system: _____. Cooling: _____. Exhaust: _____. Transmission and driveshaft: _____. Brakes and gauges: _____. Total, honest: _____. Compare it to the crate-plus-control-pack quote and to restoring the car with its original drivetrain before Chapter 1's decision is final — that comparison is the decision.

Recommended for this restoration

Shop exact-fit supplies on Amazon

- Coyote swap 1965-1970 Mustang motor mount kit
- Ford Performance Coyote control pack
- Mustang II front suspension conversion 1965-1970 Mustang
- Coyote swap oil pan classic Mustang

As an Amazon Associate, RestorationEssentials earns from qualifying purchases made through these links.

Sources Consulted

Every specification in this guide was checked against published references before printing. Where sources disagree, this guide says so in the text rather than picking a number silently. The references below are the backbone of the technical chapters:

- **Ford Performance control-pack listings (M-6017 series).** generation- and transmission-specific control packs: published contents (calibrated PCM, electronic throttle pedal, control-pack harness, OBD-II port, power distribution module, air box and inlet tube, MAF sensor, radiator hoses, HEGO sensors), the return-style fuel requirement, the Mustang-GT-only application excluding F-150 engines, and the modified/supercharged custom-calibration note, as published by Ford Performance retailers
- **Ford Coyote generation data.** Mustang GT ratings and generation changes as published (Gen 1 412–420 hp, Gen 2 435 hp, Gen 3 460 hp with direct-plus-port injection and 12.0:1 compression) via Ford Performance specification sheets and published generation tables
- **Hot Rod Coyote swap guide (MotorTrend).** swap-practice coverage: shock-tower deletion vs trimming schools, electric-fan requirement, cooling fill/surge-tank and heater-hose purge discipline, PATS anti-theft



behavior and the control-pack default, front-ump swap pan practice, and cast street-rod manifold availability

- **Mustang II / aftermarket front-suspension coverage.** tower-removal front conversions: rack-and-pinion conversion, geometry and ride-height effects, and the engine-room consequence, as published
- **Ford Mustang platform documentation.** 1965–70 unibody construction, shock-tower layout, and tunnel packaging for the Toploader/C4 era

A note on numbers. Factory literature, period road tests, and marque historians do not always agree — production totals, horsepower ratings, and paint codes all have documented conflicts. This guide uses the best-supported figure in each case and flags the conflict where one exists, so you can judge for yourself.

Emissions legality varies by jurisdiction, and the control-pack listings say so themselves: these calibrations are sold for competition/off-highway use. This guide explains what gets asked in your state; it cannot guarantee any conversion passes anywhere. Structural rust repair belongs to the restoration guides and to body shops; forced-induction Coyote builds beyond the factory-supercharged engines are outside this guide's scope.



Thank You

Thank you for choosing this RestorationEssentials guide. Whether this is your first Coyote Conversion — Mustang 1965–1970 or your fiftieth, you picked one of the great ones, and the fact that you are doing your homework before you turn a wrench puts you ahead of most of the field.

These cars survived because people like you refused to let them disappear. Every correct fastener, every properly decoded tag, every hour spent getting the details right keeps the history honest for the next owner and the next generation of enthusiasts.

A wide modern engine in a narrow classic bay — decided on paper before it was decided with a cutting wheel. Measure on your car, write the permanent changes in the worksheet, and finish it: a finished swap beats a perfect plan every Saturday night.

If this guide helped your build, we would love to hear about it. Good luck with the restoration, and enjoy every mile.

— The RestorationEssentials Team